

High-Accuracy Calibration of MIMO Radar Based on Directivity Error Model with Kronecker-Product Constraint

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Abstract—In this paper, we propose a calibration method for MIMO radar that achieves high accuracy with a small amount of calibration data by exploiting the structural properties of MIMO systems. In MIMO radar, distortions in the directional characteristics of each antenna element can adversely affect the accuracy of target estimation. Therefore, it is imperative to remove the influence of antenna directivity through calibration in order to achieve accurate and stable target estimation. Conventional methods perform calibration by representing the influence of directivity on target estimation as a single error matrix. In the proposed method, we impose a constraint such that the overall system error matrix is expressed as the Kronecker product of the Tx and Rx error matrices. This constraint reduces the number of unknown coefficients in the error matrix, thereby enabling high-precision calibration with a small amount of calibration data. Through DOA estimation simulations, we show that applying the proposed method effectively improves the estimation accuracy and expands the field of view.

Index Terms—FMCW-MIMO radar, DOA estimation, Calibration, Alternating optimization

I. INTRODUCTION

Frequency-modulated continuous wave (FMCW) multiple-input multiple-output (MIMO) radar has been widely used for applications such as obstacle detection for autonomous driving and gesture recognition with enhanced privacy. With its expanding adoption, radar modules capable of wide-angle and high-accuracy position estimation are increasingly required.

Antenna elements with high gain and wide beamwidths are required to achieve a wide field of view (FoV). When patch antennas are used, an H-plane array configuration, where the feed direction is parallel to the scanning direction, can more readily achieve wide-angle and high-gain performance than an E-plane configuration, in which the two directions are orthogonal [1][2].

However, wide-angle, high-gain antenna elements increase mutual coupling, leading to radiation pattern distortions. These distortions degrade received signals and reduce direction-of-arrival (DOA) estimation accuracy. Therefore, wide-FoV radar systems require calibration techniques to compensate for such distortions [3]–[6].

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In the conventional method [4], radiation pattern distortions are modeled by a single error matrix, which is estimated using the orthogonality between steering vectors and the noise subspace. However, the minimum number of calibration data points required for unique estimation is approximately proportional to the product of the numbers of transmit antennas (Tx) and receive antennas (Rx), resulting in rapidly increasing calibration costs for large arrays. Reducing this requirement would improve efficiency and enable accurate calibration even with limited data, such as in embedded radar modules.

In this paper, we propose a calibration method that achieves accurate calibration with few data points by imposing structural constraints on the error matrix. In MIMO radar systems, the error matrix can be modeled as a Kronecker product of two smaller matrices [7]–[9], which reduces the number of unknown parameters and the required calibration data compared to the conventional methods. Through DOA estimation simulations using the Annihilating Filter method [10], we show that the proposed method can execute array calibration accurately in the case that the number of data points is too small to calibrate by the conventional method.

II. PRELIMINARIES

A. Signal Model for MIMO Radar

In this section, we introduce the signal model for DOA estimation by MIMO radar [7]. We consider a scenario in which a MIMO radar with M_{Tx} Txs and M_{Rx} Rxs (an $M_{Tx} \times M_{Rx}$ MIMO radar) detects the angular directions of K targets. Fig. 1 shows a schematic diagram of a 2×3 MIMO radar configuration. For $i = 1, \dots, K$, let θ_i denote the direction of the i th target, and let $s_i(t)$ denote the signal waveform of the echo reflected from direction θ_i at time t . The transmitted waveforms from the Txs are assumed to be mutually orthogonal, such that inter-transmitter interference can be neglected. Furthermore, it is assumed that the distances between the Txs and Rxs are sufficiently large, and thus mutual coupling between the Txs and Rxs can be ignored. Letting $M = M_{Tx}M_{Rx}$, the signal model of the $M_{Tx} \times M_{Rx}$ MIMO

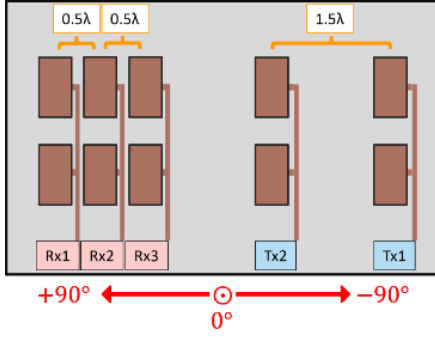


Fig. 1: Schematic diagram of 2×3 MIMO radar structure

radar becomes equivalent to that of a virtual linear array with M elements and can be expressed as

$$\mathbf{x}(t) = \sum_{i=1}^K \mathbf{G}(\theta_i) \mathbf{a}(\theta_i) s_i(t) + \mathbf{n}(t). \quad (1)$$

Here, $\mathbf{x}(t) \in \mathbb{C}^{M \times 1}$ denotes the received signal of the virtual M -element array at time t . $\mathbf{G}(\theta_i) \in \mathbb{C}^{M \times M}$ is the error matrix corresponding to direction θ_i , which represents amplitude and phase errors at each element as well as the effects of electromagnetic re-radiation among array elements. The vector $\mathbf{a}(\theta_i) \in \mathbb{C}^{M \times 1}$ is the steering vector for direction θ_i , and $\mathbf{n}(t) \in \mathbb{C}^{M \times 1}$ denotes additive white Gaussian noise at time t . The error matrix $\mathbf{G}(\theta_i)$ and the steering vector $\mathbf{a}(\theta_i)$ can each be decomposed into the Kronecker product of two matrices or vectors as follows:

$$\mathbf{G}(\theta_i) = \mathbf{G}_{\text{Tx}}(\theta_i) \otimes \mathbf{G}_{\text{Rx}}(\theta_i), \quad (2)$$

$$\mathbf{a}(\theta_i) = \mathbf{a}_{\text{Tx}}(\theta_i) \otimes \mathbf{a}_{\text{Rx}}(\theta_i). \quad (3)$$

Here, $\mathbf{G}_{\text{Tx}}(\theta_i) \in \mathbb{C}^{M_{\text{Tx}} \times M_{\text{Tx}}}$ and $\mathbf{G}_{\text{Rx}}(\theta_i) \in \mathbb{C}^{M_{\text{Rx}} \times M_{\text{Rx}}}$ are the error matrices for the Tx's and Rx's, respectively, while $\mathbf{a}_{\text{Tx}}(\theta_i) \in \mathbb{C}^{M_{\text{Tx}} \times 1}$ and $\mathbf{a}_{\text{Rx}}(\theta_i) \in \mathbb{C}^{M_{\text{Rx}} \times 1}$ are the corresponding steering vectors. The operator $\otimes$ denotes the Kronecker product.

B. Conventional Calibration Method

In this section, we introduce the conventional method [4] arranged for general cases, that is, any structure restriction is not imposed on the error matrix, and M can be any positive integer. In [4], the decomposition of the error matrix into mutual coupling matrix and gain/phase matrix is assumed, and only the case of $M = 4$ is considered.

In the conventional approach, P sets of received data are prepared, each corresponding to a single incident wave arriving from a known direction, and these data are used to estimate the error matrix. Although the signal model in (1) formulates the error matrix as being angle-dependent, the target of estimation in the calibration considered in this paper is a single angle-independent error matrix $\tilde{\mathbf{G}} \in \mathbb{C}^{M \times M}$. That is, it is assumed that variations in the error matrix with respect to the direction of arrival are sufficiently small.

For $p = 1, \dots, P$, let θ_p denote the direction of arrival for the p th calibration dataset. The received signal $\mathbf{x}_p(t)$ for the p th calibration dataset can then be expressed as

$$\mathbf{x}_p(t) = \mathbf{G}(\theta_p) \mathbf{a}(\theta_p) s(t) + \mathbf{n}(t). \quad (4)$$

Here, $s(t)$ denotes the waveform of the incident signal. For the received signal $\mathbf{x}_p(t)$ given in (4), let $\mathbf{R}_{xx,p} \triangleq \mathbb{E}[\mathbf{x}_p(t) \mathbf{x}_p^H(t)]$ be the covariance matrix, and let $\mathbf{e}_{1,p}, \dots, \mathbf{e}_{M,p}$ denote its eigenvectors ordered according to descending absolute values of their corresponding eigenvalues. Here, $\mathbb{E}[\cdot]$ denotes the expectation operator. From the orthogonality between the signal subspace and the noise subspace, the following condition is obtained:

$$\mathbf{e}_{m,p}^H \tilde{\mathbf{G}} \mathbf{a}(\theta_p) = 0 \quad \text{for } m = 2, \dots, M, \quad p = 1, \dots, P. \quad (5)$$

The condition in (5) yields $P(M_{\text{Tx}}M_{\text{Rx}} - 1)$ equations. On the other hand, since the $(1,1)$ element of the error matrix $\tilde{\mathbf{G}}$ can be fixed to unity, the number of unknown parameters contained in $\tilde{\mathbf{G}}$ is $(M_{\text{Tx}}M_{\text{Rx}})^2 - 1$. Therefore, in order to uniquely determine $\tilde{\mathbf{G}}$, the number of data points P must satisfy

$$P \geq M_{\text{Tx}}M_{\text{Rx}} - 1. \quad (6)$$

III. PROPOSED METHOD

In this section, we propose a calibration method that imposes a constraint similar to (2), namely a Kronecker product constraint, on the estimated error matrix. Section III-A formulates the Kronecker product constraint and derives transformed orthogonality conditions between the steering vectors and the noise subspace. In Section III-B, an algorithm for estimating the error matrix based on the derived conditions using alternating optimization is presented.

A. Formulation of the Kronecker Product Constraint

In conventional calibration methods, no structural constraint is imposed on the estimated error matrix $\tilde{\mathbf{G}}$. However, for models that assume MIMO radar signal processing, as described in Section II-A, it is reasonable to impose a constraint such that the estimated error matrix has a structure similar to that in (2). Moreover, the structural constraint in (2) reduces the number of unknown parameters in the estimated error matrix, thereby enabling calibration with fewer calibration data points.

The constraint imposed on the estimated error matrix is given by

$$\tilde{\mathbf{G}} = \tilde{\mathbf{G}}_{\text{Tx}} \otimes \tilde{\mathbf{G}}_{\text{Rx}}, \quad (7)$$

where $\tilde{\mathbf{G}}_{\text{Tx}} \in \mathbb{C}^{M_{\text{Tx}} \times M_{\text{Tx}}}$ and $\tilde{\mathbf{G}}_{\text{Rx}} \in \mathbb{C}^{M_{\text{Rx}} \times M_{\text{Rx}}}$ denote the estimated error matrices of the Tx's and Rx's, respectively. Substituting the constraint (7) into (5) yields the following condition:

$$\mathbf{e}_{m,p}^H [\tilde{\mathbf{G}}_{\text{Tx}} \otimes \tilde{\mathbf{G}}_{\text{Rx}}] \mathbf{a}(\theta_p) = 0 \quad \text{for } m = 2, \dots, M. \quad (8)$$

For a vector $\mathbf{b} \in \mathbb{C}^{M_{\text{Tx}} M_{\text{Rx}} \times 1}$, we define the matricization operator $\text{mat} : \mathbb{C}^{M_{\text{Tx}} M_{\text{Rx}} \times 1} \rightarrow \mathbb{C}^{M_{\text{Tx}} \times M_{\text{Rx}}}$ as

$$\text{mat}(\mathbf{b}) = [\mathbf{b}_1, \dots, \mathbf{b}_{M_{\text{Tx}}}]^T, \quad (9)$$

$$\mathbf{b}_i = [b_{(i-1)M_{\text{Rx}}+1}, \dots, b_{(i-1)M_{\text{Rx}}+M_{\text{Rx}}}]^T \quad \text{for } i = 1, \dots, M_{\text{Tx}}, \quad (10)$$

where b_x denotes the x -th element of vector $\mathbf{b}$. Since $\mathbf{e}_{m,p} \in \mathbb{C}^{M \times 1}$ and $\mathbf{a}(\theta_p) \in \mathbb{C}^{M \times 1}$ with $M = M_{\text{Tx}} M_{\text{Rx}}$, the matrices $\text{mat}(\mathbf{e}_{m,p})$ and $\text{mat}(\mathbf{a}(\theta_p))$ are of size $M_{\text{Tx}} \times M_{\text{Rx}}$. We further define $\tilde{\mathbf{g}}_{\text{Tx}} \in \mathbb{C}^{M_{\text{Tx}} \times 1}$ and $\tilde{\mathbf{g}}_{\text{Rx}} \in \mathbb{C}^{M_{\text{Rx}} \times 1}$ as

$$\tilde{\mathbf{g}}_{\text{Tx}} \triangleq [\tilde{g}_{(\text{Tx})1}, \dots, \tilde{g}_{(\text{Tx})M_{\text{Tx}}}]^T, \quad (11)$$

$$\tilde{g}_{(\text{Tx})m} \triangleq [\tilde{g}_{(\text{Tx})m1}, \dots, \tilde{g}_{(\text{Tx})mM_{\text{Tx}}}] \quad \text{for } m = 1, \dots, M_{\text{Tx}}, \quad (12)$$

$$\tilde{\mathbf{g}}_{\text{Rx}} \triangleq [\tilde{g}_{(\text{Rx})1}, \dots, \tilde{g}_{(\text{Rx})M_{\text{Rx}}}]^T, \quad (13)$$

$$\tilde{g}_{(\text{Rx})m} \triangleq [\tilde{g}_{(\text{Rx})m1}, \dots, \tilde{g}_{(\text{Rx})mM_{\text{Rx}}}] \quad \text{for } m = 1, \dots, M_{\text{Rx}}. \quad (14)$$

Here, $\tilde{g}_{(\text{Tx})xy}$ and $\tilde{g}_{(\text{Rx})xy}$ denote the (x, y) -th elements of $\tilde{\mathbf{G}}_{\text{Tx}}$ and $\tilde{\mathbf{G}}_{\text{Rx}}$, respectively. Using the matricization operator and (11)–(13), the condition (8) can be rewritten as

$$\tilde{\mathbf{g}}_{\text{Tx}}^T [\text{mat}(\mathbf{e}_{m,p}^*) \otimes \text{mat}(\mathbf{a}(\theta_p))] \tilde{\mathbf{g}}_{\text{Rx}} = 0 \quad \text{for } m = 2, \dots, M. \quad (15)$$

Since the right-hand side is a scalar, transposing the left-hand side yields

$$\tilde{\mathbf{g}}_{\text{Rx}}^T [\text{mat}(\mathbf{e}_{m,p}^*) \otimes \text{mat}(\mathbf{a}(\theta_p))]^T \tilde{\mathbf{g}}_{\text{Tx}} = 0 \quad \text{for } m = 2, \dots, M. \quad (16)$$

B. Proposed Calibration Algorithm

The conditions (15) and (16) are expressed as products of two unknown vectors, $\tilde{\mathbf{g}}_{\text{Tx}}$ and $\tilde{\mathbf{g}}_{\text{Rx}}$. Therefore, it is difficult to determine $\tilde{\mathbf{G}}_{\text{Tx}}$ and $\tilde{\mathbf{G}}_{\text{Rx}}$ simultaneously. In the proposed method, these matrices are estimated via alternating optimization.

$\tilde{\mathbf{g}}_{\text{Rx}}$ can be estimated with known $\tilde{\mathbf{g}}_{\text{Tx}}$ by the following procedure. We define $\mathbf{F}_{\text{Rx}}$ as

$$\mathbf{F}_{\text{Rx}} \triangleq [\mathbf{H}_{(2,1)}^T \tilde{\mathbf{g}}_{\text{Tx}}, \dots, \mathbf{H}_{(M,P)}^T \tilde{\mathbf{g}}_{\text{Tx}}]^T, \quad (17)$$

where

$$\mathbf{H}_{(m,p)} \triangleq \text{mat}(\mathbf{e}_{m,p}^*) \otimes \text{mat}(\mathbf{a}(\theta_p)), \quad (18)$$

for $m = 2, \dots, M$ and $p = 1, \dots, P$. Let the first element of $\tilde{\mathbf{g}}_{\text{Rx}}$ be 1, and the remaining vector be $\tilde{\mathbf{g}}'_{\text{Rx}}$. Similarly, let the first column of $\mathbf{F}_{\text{Rx}}$ be $\mathbf{f}_{\text{Rx}}$, and the remaining part be $\mathbf{F}'_{\text{Rx}}$. Then, $\tilde{\mathbf{g}}'_{\text{Rx}}$ can be obtained as the solution that minimizes $\|\mathbf{F}'_{\text{Rx}} \tilde{\mathbf{g}}_{\text{Rx}}\|$, given by

$$\tilde{\mathbf{g}}'_{\text{Rx}} = -(\mathbf{F}_{\text{Rx}}^H \mathbf{F}'_{\text{Rx}})^{-1} \mathbf{F}_{\text{Rx}}^H \mathbf{f}_{\text{Rx}}. \quad (19)$$

From $\tilde{\mathbf{g}}'_{\text{Rx}}$, the elements of $\tilde{\mathbf{g}}_{\text{Rx}}$ and $\tilde{\mathbf{G}}_{\text{Rx}}$ are obtained.

The formula to calculate $\tilde{\mathbf{g}}_{\text{Tx}}$ with known $\tilde{\mathbf{g}}_{\text{Rx}}$ is obtained in the same manner. We define $\mathbf{F}_{\text{Tx}}$ as

$$\mathbf{F}_{\text{Tx}} \triangleq [\mathbf{H}_{(2,1)} \tilde{\mathbf{g}}_{\text{Rx}}, \dots, \mathbf{H}_{(M,P)} \tilde{\mathbf{g}}_{\text{Rx}}]^T. \quad (20)$$

Algorithm 1 Error Matrix Estimation with Kronecker Product Constraint

Input: $\mathbf{H}_{(m,p)}$ for $m = 2, \dots, M$, $p = 1, \dots, P$

- 1: **Initialize** $\tilde{\mathbf{G}}_{\text{Tx}}^{(0)} \leftarrow \mathbf{I}_{M_{\text{Tx}}}$, $\tilde{\mathbf{G}}_{\text{Rx}}^{(0)} \leftarrow \mathbf{I}_{M_{\text{Rx}}}$, $\tilde{\mathbf{G}}^{(0)} \leftarrow \tilde{\mathbf{G}}_{\text{Tx}}^{(0)} \otimes \tilde{\mathbf{G}}_{\text{Rx}}^{(0)}$
- 2: Calculate $\tilde{\mathbf{g}}_{\text{Tx}}^{(0)}$ using $\tilde{\mathbf{G}}_{\text{Tx}}^{(0)}$ based on (11)
- 3: **for** $i = 1, \dots, i_{\text{max}}$ **do**
- 4: Calculate $F_{\text{Rx}}^{(i)}$ using $\tilde{\mathbf{g}}_{\text{Tx}}^{(i-1)}$ based on (17)
- 5: Calculate $\tilde{\mathbf{g}}_{\text{Rx}}^{(i)}$ using $F_{\text{Rx}}^{(i)}$ based on (19)
- 6: Update $\tilde{\mathbf{G}}_{\text{Rx}}^{(i)}$ using $\tilde{\mathbf{g}}_{\text{Rx}}^{(i)}$
- 7: Calculate $F_{\text{Tx}}^{(i)}$ using $\tilde{\mathbf{g}}_{\text{Rx}}^{(i)}$ based on (20)
- 8: Calculate $\tilde{\mathbf{g}}_{\text{Tx}}^{(i)}$ using $F_{\text{Tx}}^{(i)}$ based on (21)
- 9: Update $\tilde{\mathbf{G}}_{\text{Tx}}^{(i)}$ using $\tilde{\mathbf{g}}_{\text{Tx}}^{(i)}$
- 10: Update $\tilde{\mathbf{G}}^{(i)} \leftarrow \tilde{\mathbf{G}}_{\text{Tx}}^{(i)} \otimes \tilde{\mathbf{G}}_{\text{Rx}}^{(i)}$
- 11: **if** $\|\tilde{\mathbf{G}}^{(i)} - \tilde{\mathbf{G}}^{(i-1)}\|_{\text{F}} \leq \alpha$ **then**
- 12: **break**
- 13: **end if**
- 14: **end for**
- 15: **return** $\tilde{\mathbf{G}}$

Let the first element of $\tilde{\mathbf{g}}_{\text{Tx}}$ be 1, and the remaining vector be $\tilde{\mathbf{g}}'_{\text{Tx}}$. Similarly, let the first column of $\mathbf{F}_{\text{Tx}}$ be $\mathbf{f}_{\text{Tx}}$, and the remaining part be $\mathbf{F}'_{\text{Tx}}$. Then, $\tilde{\mathbf{g}}'_{\text{Tx}}$ can be obtained as the solution that minimizes $\|\mathbf{F}'_{\text{Tx}} \tilde{\mathbf{g}}_{\text{Tx}}\|$, given by

$$\tilde{\mathbf{g}}'_{\text{Tx}} = -(\mathbf{F}_{\text{Tx}}^H \mathbf{F}'_{\text{Tx}})^{-1} \mathbf{F}_{\text{Tx}}^H \mathbf{f}_{\text{Tx}}. \quad (21)$$

From $\tilde{\mathbf{g}}'_{\text{Tx}}$, the full vector $\tilde{\mathbf{g}}_{\text{Tx}}$ and all elements of $\tilde{\mathbf{G}}_{\text{Tx}}$ can be reconstructed.

Algorithm 1 summarizes the proposed estimation procedure with the Kronecker product constraint. Here, i_{max} denotes the maximum number of iterations, and α is the convergence threshold. At each iteration, $\tilde{\mathbf{G}}_{\text{Rx}}$ and $\tilde{\mathbf{G}}_{\text{Tx}}$ are alternately updated, and the iteration terminates when the Frobenius norm of the difference between successive estimates of $\tilde{\mathbf{G}}$ falls below α .

The number of equations obtained from P calibration data points is $P(M_{\text{Tx}} M_{\text{Rx}} - 1)$, identical to that of the conventional method. On the other hand, the total number of unknown parameters in $\tilde{\mathbf{G}}$ is $M_{\text{Tx}}^2 + M_{\text{Rx}}^2 - 2$. Therefore, the lower bound on the required number of calibration data points is given by

$$P \geq \left\lceil \frac{M_{\text{Tx}}^2 + M_{\text{Rx}}^2 - 2}{M_{\text{Tx}} M_{\text{Rx}} - 1} \right\rceil, \quad (22)$$

where $\lceil \cdot \rceil$ denotes the ceiling operator. From (22), the lower bound on the required number of calibration data points for the proposed method is $O(M_{\text{Tx}} + M_{\text{Rx}})$, while that for the conventional method is $O(M_{\text{Tx}} M_{\text{Rx}})$.

The calibration method proposed in this paper assumes that the mutual coupling between the Tx's and Rx's is negligible. Therefore, it is desirable for Tx's and Rx's to be well-separated. Accordingly, the simulations in this paper employ the MIMO radar structure shown in Fig. 2. The MIMO radar illustrated

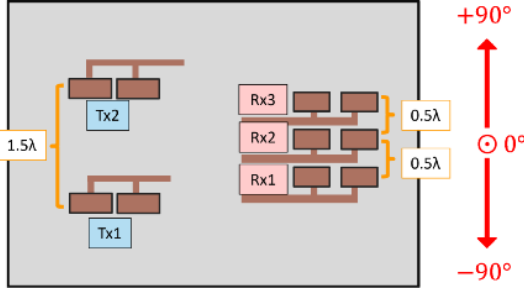


Fig. 2: The MIMO radar to be calibrated

TABLE I: Simulation parameters

For calibration	
# of sources	1
DOA	$-75^\circ, -50^\circ, -25^\circ, 0^\circ, 25^\circ, 50^\circ, 75^\circ$
# of data	7
SNR	20 dB
# of snapshots	1
α	10^{-6}
$i_{\max}$	100
For DOA estimation	
# of sources	1
DOA of 1st source	$-90^\circ \sim 90^\circ, 1^\circ$ interval
SNR	10 dB
# of snapshots	1
DOA estimator	Annihilating Filter [10]
# of trials	100
Evaluator	RMSE

in Fig. 2 is designed to reduce mutual coupling between the Txs and Rxes by appropriately arranging their positions.

IV. SIMULATION

In this section, the effectiveness of the proposed calibration method is evaluated through simulations of calibration and DOA estimation.

The 2×3 MIMO radar shown in Fig. 2 is used as the radar to be calibrated. In signal processing, it is treated as a virtual array equivalent to a six-element Uniform Linear Array (ULA). The upper half of TABLE I summarizes the simulation parameters for calibration. The number of calibration data points is equal to the minimum required number of calibration data points for the conventional method when $M_{\text{Tx}} = 2$ and $M_{\text{Rx}} = 3$. The lower half of TABLE I lists the simulation parameters for DOA estimation. The number of snapshots is set to 1 since we assumed signal processing for MIMO radar signals. The Annihilating Filter method [10] is employed as the DOA estimator due to its low computational complexity and good performance for signals with a small number of snapshots. The estimated error matrix is applied by left-multiplying the inverse of the estimated error matrix to the received signal of the MIMO radar. In other words, the relationship between the received signal before calibration $\mathbf{x}(t)$ and after calibration $\mathbf{x}'(t)$ is given by

$$\mathbf{x}'(t) = \tilde{\mathbf{G}}^{-1} \mathbf{x}(t). \quad (23)$$

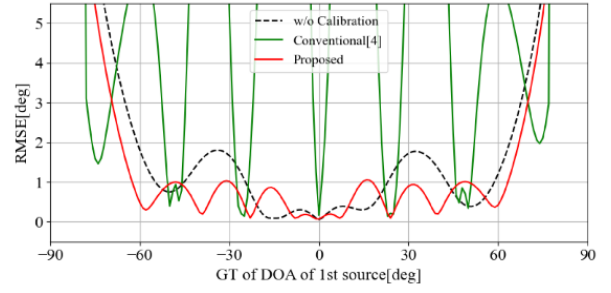


Fig. 3: Relationship between the ground-truth DOA of the first source and RMSE

The evaluation metric, Root-Mean-Squared Error (RMSE), is defined as

$$\text{RMSE}(\theta_{\text{GT}}) \triangleq \sqrt{\frac{1}{N_T K} \sum_{q=1}^{N_T} \sum_{i=1}^K (\theta_{\text{GT},i} - \theta_{\text{est},i,q})^2}, \quad (24)$$

where $\theta_{\text{GT},i}$ denotes the ground-truth DOA of the i -th source, $\theta_{\text{est},i,q}$ denotes the estimated DOA of the i -th source in the q -th trial, K is the number of sources, and N_T is the number of trials.

Fig. 3 shows the relationship between the ground-truth DOA and the RMSE. In Fig. 3, the black plots indicate the estimation accuracy without calibration, the green plots correspond to the conventional calibration method, and the red plots represent the proposed calibration method.

Compared with the conventional method, which exhibits large fluctuations and unstable performance, the proposed method maintains stable accuracy of approximately 1° or less over a DOA range of about $\pm 65^\circ$. Furthermore, focusing on the RMSE rise near $\pm 60^\circ$ (side directions), the proposed method improves estimation accuracy compared with the case without calibration. This indicates that the proposed method effectively expands the FoV.

V. CONCLUSION

This paper proposes a calibration method that imposes a Kronecker product constraint on the estimated error matrix based on MIMO extended signal processing. By reducing the number of unknown parameters, the proposed method enables high-accuracy calibration with a small number of calibration data points.

Simulation results demonstrated that the proposed method significantly improves DOA estimation accuracy over a wide angular range, even in scenarios where conventional methods fail due to insufficient calibration data. Furthermore, the effective FoV is expanded compared to the uncalibrated case.

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